DHTMLX Gantt Chart

Evaluation & Implementation Recommendation

Prepared for: ResourceVue V2 Platform Team

June 2025

Executive Summary

This document presents a technical and strategic evaluation of DHTMLX Gantt as the Gantt chart and Critical Path Analysis (CPA) component for the ResourceVue V2 platform. The assessment covers feature fit, integration approach, licensing model, a comparison against alternative tools, and a final recommendation.

1. Features We Can Build with DHTMLX Gantt

DHTMLX Gantt supports the full ResourceVue data hierarchy — **Project** → **Work Packages (WRs)** → **Tasks** → **Milestones** — natively and with minimal custom development. The following capabilities are available out of the box:

- Hierarchical Gantt view with collapsible Project / WR / Task levels
- Visual Critical Path highlighting (built-in plugin on Pro/Enterprise tier)
- Drag-and-drop rescheduling with live two-way sync to your backend
- Inline editing of task names, start/end dates, durations, and progress
- Milestone display — connected or standalone
- Zoom levels: day, week, and month views
- Custom task types and per-type visual styling (WR vs. Task vs. Milestone)
- Role-based read/edit controls configurable at any task level
- Real-time synchronisation via REST API or WebSockets

2. Licensing Model

DHTMLX Gantt offers a **one-time perpetual licence** — there are no recurring subscription fees. Once purchased, the licence version is owned indefinitely. Optional support and update renewals are available annually but are not mandatory.

Key licensing tiers relevant to this evaluation:

Tier	Developers	Projects	SaaS Use	Critical Path	Exports

Community (GPL)	Unlimited	Unlimited	✓ (with attribution)	✗	✗
Enterprise	Up to 20	Up to 5	✓	✓	Add-on
Ultimate	Unlimited	Unlimited	✓	✓	Included

Table 1 — DHTMLX Gantt licence tiers (one-time perpetual, no subscription)

For a customer-facing SaaS platform with Critical Path requirements, the **Enterprise tier** is the minimum viable option. The **Ultimate tier** removes all developer and project limits and includes export modules.

> *Source code is included with the Ultimate licence, enabling full in-house maintenance even if the vendor were to cease operations.*

3. Inline Editing Support

DHTMLX Gantt provides full inline editing. Users can click directly on any task bar or grid cell to edit task name, start date, end date, duration, and progress — without opening a separate modal or form. All edit events are captured by the library's event system and forwarded to the backend via REST.

Key inline editing capabilities:

- Click-to-edit on task bars and the left-hand grid simultaneously
- Drag-and-drop to reschedule tasks or adjust duration
- Lightbox (optional pop-up form) for extended field editing
- Granular readonly controls per task, per user role, or per task type
- Events fired on every change: *onAfterTaskUpdate*, *onAfterTaskDrag*, *onAfterLinkAdd*, etc.

4. Creating & Editing Projects, Work Packages, and Tasks

The library accepts a standard JSON structure where each item carries an ID, label, start date, duration, progress value, and parent reference. This maps directly to the ResourceVue hierarchy:

Level	DHTMLX Type	Parent Reference	Notes
Project	project	0 (root)	Top-level grouping node
Work Package	task / WR	Project ID	Custom type; styled distinctly
Task	task	WR or Project ID	Leaf-level work item
Milestone	milestone	Project or WR ID	Zero-duration; connected or standalone

Table 2 — ResourceVue hierarchy mapped to DHTMLX task types

Both start and end dates are supported. End date can be provided explicitly or auto-calculated from start date and duration. All fields are editable inline or via the lightbox form.

5. Critical Path — Capabilities & Limitations

✓ What works out of the box

- Built-in Critical Path plugin — enabled with two lines of configuration
- Automatically traverses the dependency graph and highlights the longest chain
- Critical tasks rendered in a distinct colour (configurable)
- Multiple critical paths supported simultaneously
- Recalculated in real time as tasks are edited or rescheduled

■ Important constraint to understand

Critical path calculation is driven entirely by the **dependency links** defined between tasks — not inferred from the parent/child hierarchy. The parent field controls visual grouping only.

This means: if the existing ResourceVue data model does not already store task-to-task and task-to-milestone dependency relationships, a **one-time data-mapping effort** is required to generate those links before the critical path can function correctly. This is a defined, bounded piece of work — not an ongoing limitation.

6. Plug-and-Play vs. Development Effort

DHTMLX Gantt is a **JavaScript UI library**, not a hosted SaaS tool. It requires integration work, but the scope is well-defined and low-risk.

Work Stream	Effort Estimate	Notes
Chart integration into portal	1–2 weeks	React component wrapper, layout, theme
BFF / facade API layer	1 week	Aggregates Project/WR/Task data into DHTMLX JSON format
Two-way sync (edit → backend)	1 week	REST event hooks, optimistic updates
Dependency mapping (for CPA)	1 week	One-time: map existing links to DHTMLX format
Styling & UX polish	0.5–1 week	Per-type colours, tooltips, zoom controls

Table 3 — Estimated implementation effort (working days per stream)

Total estimated effort: 4–6 weeks for a production-ready, fully integrated Gantt with critical path. This assumes the portal's REST APIs are already in place.

7. DHTMLX Licence vs. Building a Custom Solution

Criteria	DHTMLX Licence	Custom Build
Time to first working Gantt	1–2 weeks	2–4 months
Critical path implementation	Built-in plugin	Complex algorithm; 4–8 weeks alone
Inline editing & drag-drop	Included	Must build from scratch
Ongoing maintenance	Low — source code owned	High — full team responsibility

Reuse across projects	Enterprise: 5 projects; Ultimate: unlimited	Per-project rebuild cost
Vendor dependency risk	Low — source code in-house after purchase	None
Long-term flexibility	Full — source code modifiable	Full

Table 4 — DHTMLX licence vs. custom build comparison

A custom Gantt with production-grade critical path analysis is a substantial engineering undertaking. The DHTMLX licence provides a battle-tested foundation while retaining full source code ownership, significantly reducing time-to-market and implementation risk.

8. Alternative Tools — Comparison

Tool	Licence Model	Critical Path	React	Summary
DHTMLX Gantt	One-time perpetual	✓ (Pro+)	✓	Most complete feature set; best architectural fit
Syncfusion Gantt	Annual subscription	✓	✓	Strong UI; recurring cost is a long-term consideration
Bryntum Gantt	One-time perpetual	✓	✓	Feature-comparable to DHTMLX; slightly higher entry point
Frappe Gantt	Free (MIT)	✗	Community wrappers	Lightweight; no CPA; limited for complex hierarchies
vis-timeline	Free (MIT)	✗	Community wrappers	Good for simple timelines; not a full Gantt

Table 5 — Gantt library comparison (✓ = supported natively, ✗ = not supported)

Of the alternatives, **Bryntum Gantt** is the closest commercial equivalent and worth a parallel evaluation if a second option is desired. Free options (Frappe Gantt, vis-timeline) lack native critical path support and are not suitable for this use case without significant custom development.

9. Recommendation

Recommended: Proceed with DHTMLX Gantt — Enterprise or Ultimate Tier

Based on the evaluation, DHTMLX Gantt is the recommended path forward for the following reasons:

- **Architectural fit:** React frontend, .NET microservices, and REST APIs are fully compatible — DHTMLX is frontend-decoupled by design.
- **Feature completeness:** Hierarchical tasks, inline editing, drag-and-drop, and critical path are all included without custom development.

- **Ownership:** One-time perpetual licence with source code eliminates vendor lock-in and protects the platform long-term.
- **Reusability:** The Enterprise tier covers up to 5 projects; the Ultimate tier covers unlimited projects — making the investment reusable across the product portfolio.
- **Implementation timeline:** A working production integration is achievable in 4–6 weeks, versus several months for a custom build.
- **Risk profile:** The library is mature, well-documented, and widely used in enterprise SaaS products.

Suggested next step: Run the 30-day free Pro trial against a real slice of ResourceVue data (Project → WR → Task → Milestone) to validate the integration and confirm the critical path renders as expected — before any licence is purchased. This de-risks the decision at zero cost.
